

Object Oriented Programming Lab

Lab 05

Instructions

Work on this lab individually. You can use your books, notes, handouts etc but you are not allowed to borrow anything from your peer student.

Write down the lab number, task number, your name and roll number at the start of each task code using comments.

For Example

```

/*****
Lab:    xx
Task:   yy
Developer:
        Roll #: XXXYYYZ000
        Name:  XYZ
*****/

```

What you have to do

Program the following tasks in C++, compile and execute them.

Task

Write a Circle class that has the following member variables:

- radius—a double
- pi—a double initialized with the value 3.14159

The class should have the following member functions:

- Default Constructor—a default constructor that sets radius to 0.0
- Constructor—accepts the radius of the circle as an argument
- setRadius—a mutator function for the radius variable
- getRadius—an accessor function for the radius variable
- getArea—returns the area of the circle, which is calculated as $\text{area} = \text{pi} * \text{radius} * \text{radius}$
- getDiameter—returns the diameter of the circle, which is calculated as $\text{diameter} = \text{radius} * 2$
- getCircumference—returns the circumference of the circle, which is calculated as $\text{circumference} = 2 * \text{pi} * \text{radius}$

Write a main() that demonstrates the Circle class by asking the user for the circle's radius, creating a Circle object, then reporting the circle's area, diameter, and circumference

😊😊😊 **BEST OF LUCK** 😊😊😊
